EE712 - Embedded Systems Design

Electrical Engineering IIT Bombay

Laboratory Experiment - I

Problem set: 1.2 Date: January 16, 2025

Parallel I/Os and Arithmetic Operations

Aim: To perform parallel input and output operations using the GPIO ports of the TIVA boards and perform basic arithmetic operation of the inputs and give proper to output to the LED array provided.

Components to be used:

- TIVA C Board
- Resistor Array (A471J)
- LED Array (BA10G1)

Problem Definition: Given a microarchitecture where each instruction is represented using 8 bits. Bits [7:6] defines the opcode , bits [5:3] defines the operand 1 and bits [2:0] defines operand 2.
Input to the machine will be an instruction which will be of 8 bits(lets say In[7:0])) and output from the machine will be also of 8 bits (lets say Out[7:0]).

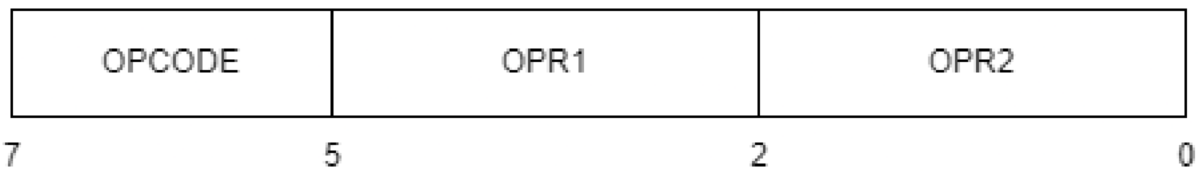

Figure 1: Instruction definition

Based on the decoded opcode, each instruction will perform a particular function which is provided in the below table.

Opcode	Operation
00	Out[7:6] <- opcode; Out[5:3] <- Opr1; Out[2:0] <- Opr2;
01	Out[7:6] <- opcode; Out[5:0] <- Opr1 + Opr2;
10	Out[7:6] <- opcode; Out[5:0] <- Opr1 - Opr2;
11	Out[7:6] <- opcode; Out[5:0] <- Opr1 * Opr2;

Table 1: Microarchitecture

GPIO Pins to be used:

- **Inputs:** *In7_In6_In5_In4_In3_In2_In1_In0 => PA7_PA6_PA5_PA4_PA3_PB5_PB4_PB3*
- **Outputs:** *Out7_Out6_Out5_Out4_Out3_Out2_Out1_Out0 => PE4_PE3_PD2_PD1_PD0_PE2_PE1_PE0*

NOTES:

1. Include following header files in your main.c
   #include "inc/hwmemmap.h" //Macros defining memory map of the device
   #include "driverlib/sysctl.h" //Prototypes for the system control driver
   #include "driverlib/gpio.h" //GPIO API

2. GPIOs on TM4C123 can be configured for 2mA, 4mA, 8mA drive strength.On reset, GPIOs are default to 2mA drive strength. While interfacing LEDs, if the LED current is more than 8mA then we cannot connect it. directly to the microcontroller pin. It might damage the chip due to high current. A good way to sink the current to the ground instead of taking into the board.

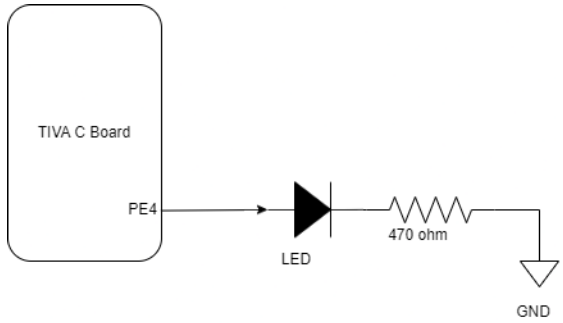

Figure 2: Connection of output LED to GPIO pin

3. 7406 buffer can also be to avoid accidental damage to microcontroller pins as an alternative.

References:

1. LED Array Datasheet : LED_Array_Datasheet

2. Resistor Array Datasheet: Resistor_Array_Datasheet